Electronic Principles and Devices · Unit_01 · Unit-1 Semiconductor Diode

DRAFT — UE26EC141A · Unit_01 · Unit-1 Semiconductor Diode · Zener and Avalanche breakdown, Temperature effects

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
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M1

Zener and Avalanche breakdown, Temperature effects

1 Zener and Avalanche breakdown, Temperature effects

Study focus: Zener and Avalanche breakdown, Temperature effects. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000030 · EV0000031

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we're going to learn about avalanche breakdown and Zener breakdown.

Key details

- Before that, we'll go for refreshing of the last topic, comparison of the diode characteristics of germanium, silicon, and gallium arsenide.
- We see that there are different voltages at which the different semiconductor device has different conducting voltages and with respect to forward bias and reverse bias.
- In the forward bias, we see the germanium, silicon, and gallium arsenide, which conduct at different voltages.
- The reverse bias also, the breakdown also occurs at different voltages.
- The first is avalanche breakdown for a normal diode we look at.
- The forward bias region is this, the reverse bias region is this.
- So in the reverse bias condition, as the voltage increases, the free charge carriers obtain velocity and associated kinetic energy and release additional carriers through collision with the other atomic structure.
- Hence, the covalent bonds are broken and electron-hole pair are generated.
- This electron will hit to the walls of an atom, as we see here, and acquire a energy from applied voltage.
- This will hit based on the energy acquired from applied voltage and produce more and more free electrons.
- This free electron again will hit the another atoms, and this cumulative process is called avalanche multiplication or ionization process.

- The Zener breakdown occurs when both P and N-type materials are heavily doped in a semiconductor diode, hence depletion region is narrowed down, as we see from the diagram here.
- When a small reverse voltage is applied across the diode, a very strong electric field is produced.
- Due to this electric field, covalent bonds breaks and produce a large number of free charge carriers.
- So coming to the comparison between the Zener breakdown and avalanche breakdown, first we start with Zener, and on the right-hand side we have avalanche.
- The first point is, the process in which the electron move across a barrier from valence band of the p-type material to the conduction band of n-type material is known as breakdown voltage, whereas the process of applying a high voltage and increasing the free electron.
- Coming to the second point, this is observed in a.
- Zener diode having a Zener breakdown, which varies from five to eight volts, whereas in the case of an avalanche breakdown, this is observed in the normal diode having a voltage greater than the eight volts.
- Number three, the valence electrons are pulled out into a conduction due to the high electric field in the narrow depletion region.
- Here, the valence electrons are pushed to the conduction due to the energy imparted with the accelerated electron which gain their velocity due to the collision with other atom.
- Coming to the fourth point, the increase in temperature decreases with the breakdown voltage.
- The increase in temperature will increase the breakdown voltage here.
- The VI characteristics of Zener diode breakdown has sharp value.
- The VI characteristics of avalanche breakdown is not sharp as.
- Zener breakdown, as we see from the diagram.
- Then in the Zener, it occurs in diode that are lightly doped.
- The effect of variation in the temperature across the semiconductor diode is observed in both forward as well as the reverse characteristics.
- The rise in temperature generates more electron-hole pair, thus conductivity increases and thus increase in the current.
- The PN junction diode parameter like reverse saturation current, bias current, reverse breakdown voltage, and barrier voltage are dependent on the temperature.
- Increase in temperature increases the carrier concentration.
- As a result, the knee voltage decreases, breakdown voltage increases, while the reverse saturation current also increases.
- This we can understand with the help of a forward bias and a reverse.

Example or application from the lecture

- Let us also consider the effect of temperature on VI characteristics of the diode.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 05:00–09:59

Under reverse bias, the change in the temperature across the diode, the reverse saturation current of the diode increases with increase in temperature.

Key details

- That is, there is a rise of 7% for every degree rise in temperature for both germanium and silicon, and approximately doubles for every 10 degree rise in the temperature.
- The reverse breakdown voltage, VR, also increases with the temperature increase.
- Starting from right to the left, initially we have -75 degree centigrade, where the cut-in voltage is at this point.
- It may be around 0.1 or even less than 0.1.
- Whereas at 125 degree, the breakdown voltage is increasing, and reverse saturation current also doubles for every 10 degree rise in the temperature.
- Determine the value if the temperature is increased by 40.
- We know that for every 10 degree rise in the temperature, the reverse saturation current double.
- Determine the values at 40 degree centigrade.
- The difference is 15 degree.
- At 40 degree, what is the current?
- We need to calculate the current in the forward bias condition if the forward bias is 0.2 at room temperature.
- If the temperature is increased to 30 degree centigrade.
- Therefore, VT is 0.8 and actual current calculation is 1.83 amps.
- Find the current at 40 degree centigrade.
- We use the whole formula last time what we used.
- Substitute that and get the ID which is 281.49 milliamps.
- Next, fourth, the reverse saturation current of germanium is two pico at 27 degree.
- So we use the same principle concept where IS calculated first.
- Substitute the value and calculate the reverse saturation current.
- Determine the forward bias also.
- The forward bias voltage VA.

Example or application from the lecture

- In the example, the reverse saturation current of silicon diode is at 20 degree is 0.1 microamp.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Zener and Avalanche breakdown, Temperature effects transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/004_Zener_and_Avalanche_breakdown_Temperature_effects__83cce28872db58b1.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Zener and Avalanche breakdown, Temperature effects transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/004_Zener_and_Avalanche_breakdown_Temperature_effects__83cce28872db58b1.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Zener and Avalanche breakdown, Temperature effects transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/004_Zener_and_Avalanche_breakdown_Temperature_effects__83cce28872db58b1.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9980; only the JSON segmentation supplies the displayed timestamps.
Zener and Avalanche breakdown, Temperature effects transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/004_Zener_and_Avalanche_breakdown_Temperature_effects__83cce28872db58b1.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.

2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00014	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0081. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0082 to s0164. Sidecar variants compare...
SRC00015	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0081. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0082 to s0164. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00016	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0081. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0082 to s0164. Sidecar variants compare...
SRC00017	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0081. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0082 to s0164. Sidecar variants compare...

C Revision and retrieval practice